

AAYUSH KUMAR

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Education

Vellore institute of technology

B.Tech in Computer Science and Engineering

Vellore, Tamil Nadu

Jun 2024 – May 2028 (Expected)

Technical Skills

Languages: Python, Go, JavaScript, C, C++, SQL

Frameworks & Libraries: Django, Flask, React.js, Next.js, Express.js, Node.js, LangGraph, RabbitMQ

Tools & Technologies: Docker, Kubernetes, Nginx, Traefik, PostgreSQL, MongoDB, Redis, Socket.io, CI/CD, Git

Experience

Loopedin

Dec 2025 – Present

Software Engineer Intern

Remote

- Built and scaled AI-agent connectors using LangGraph, enabling structured multi-step workflows such as summarization, task orchestration, and automated document generation.
- Implemented concurrent execution of agent workflows and connector pipelines, reducing end-to-end AI response time by **25–35%**.
- Designed robust integration layers with optimized state management and execution flow, improving reliability and supporting high-throughput task processing.

Yuusoft

Jul 2025 – Aug 2025

Software Engineer Intern

Remote, Singapore

- Engineered an AI-assisted automation tool using Discord + Claude to modify code, generate patches, and push Git commits via natural language, significantly improving developer workflow efficiency.
- Built core backend and platform features for Yuusoft's visual novel creator tool, including scene-management logic, input-processing pipelines, and structured data flows.
- Improved platform performance and responsiveness by optimizing state management and render paths, reducing UI latency and unnecessary re-renders by **30%** and improving overall UX reliability..

Projects

Torrcli - Terminal-Based BitTorrent Client | *Python, libtorrent, rich* | [GitHub](#)

- Built a daemon-based BitTorrent CLI client with robust session management, capable of handling multiple concurrent torrents with real-time status updates.
- Implemented end-to-end torrent workflows (search, metadata retrieval, streaming, and downloads), reducing startup and metadata fetch latency through efficient async I/O and session reuse.
- Designed a resilient terminal interface with real-time progress views, session metrics, and error-tolerant command handling for a production-grade CLI experience.

Kargo - Open-Source PaaS on Kubernetes | *Express.js, Next.js, MongoDB, Docker, Kubernetes* | [GitHub](#)

- Developed core backend services for an open-source PaaS platform that automatically deploys user-provided Docker images onto Kubernetes, building secure APIs for authentication, deployment orchestration, and lifecycle management.
- Engineered automatic Kubernetes manifest generation (Deployments, Services, Ingress, volumes, secrets), eliminating manual cluster interaction and cutting deployment setup time by **60%**.
- Improved deployment stability, scalability, and observability by adding container-status monitoring, error-resilient rollout logic, and health-check mechanisms to ensure a reliable and production-grade application.

Crate - Daemonless Container Runtime | *Go, Linux Namespaces, OCI Images* | [GitHub](#)

- Built a daemonless container runtime in Go supporting both rootless and rootful containers, with runtime state persisted directly on disk instead of a long-lived background daemon.
- Implemented container isolation using Linux namespaces (PID, UTS, Mount, User, Network), `pivot_root`-based filesystem setup, rootless networking via `pasta`, and PTY-backed interactive process execution.
- Developed OCI/Docker image support including registry pulls, manifest resolution, layer extraction, local blob/config storage, detached container lifecycle management (`start`, `stop`, `logs`, `ps`), and reference-based blob pruning.

Achievements & Extracurricular

- Google Summer of Code 2025 @AboutCode:** Contributed to the open-source ecosystem by building a high-performance file explorer for [ScanCode.io](#), an industry-grade code analysis toolkit used to detect licenses, copyrights, and vulnerabilities across large codebases
- Patent Published - Encrypted Traffic Intrusion Detection (2026):** Co-invented a multi-modal AI cybersecurity system for detecting threats in encrypted network traffic using metadata analysis and anomaly detection models.
- Volunteer - Gravitas Website Team:** Helped build the backend for VIT's annual technical fest platform, serving **65k+ unique users** and handling **1M+ page visits**.
- Winner - Cicada 3301 (CTF Competition):** Secured **1st place** in a university-wide cybersecurity challenge.
- Third Place - Coderush 2.0 (Competitive Coding):** Achieved **3rd place** in a university-wide coding competition.